



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 5 · STANDARD 6.GR.1

Area of Triangles

Flagship

Lesson 5-3-flagship · Teacher Copy

Name: _____

Date: _____

Class: _____

SKY HARBOR MISSION

The Triangular Skylight

You are the lead architect at Sky Harbor Studio. The city has hired your firm to design a glass skylight made of triangular panels for the new transit hub. Before the glass can be cut, you must master the area of triangles — every wrong measurement wastes expensive glass.



TODAY'S OBJECTIVES

Content Objective: I can find the area of a triangle using the formula $A = \frac{1}{2} \times \text{base} \times \text{height}$.

Language Objective: I can explain my steps using the words base, height, area, and perpendicular.

See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Base

Height

Area

Perpendicular

Composite figure

Formula



Tap any word to see what it means and a picture.

1 The side of a triangle used as the bottom for measuring area is the

2 The perpendicular distance from the base to the opposite vertex is the

3 The amount of flat space inside a triangle is its .

4 Two lines that meet at a 90° angle are .

5 A shape made of two or more basic shapes combined is a .

6 A math rule written with symbols, like $A = \frac{1}{2} \times b \times h$, is a .

Reflect — Exit Ticket

A triangle has a base of 11 inches and a height of 8 inches. What is its area?

- A. 44 sq in
- B. 88 sq in
- C. 19 sq in
- D. 44 in

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Base
2. **Notes 2:** Height
3. **Notes 3:** Area
4. **Notes 4:** Perpendicular
5. **Notes 5:** Composite figure
6. **Notes 6:** Formula
7. **Watch for this mistake:** *A common mistake in Area of Triangles is skipping the key idea: "Area of a triangle = $\frac{1}{2} \times \text{base} \times \text{height}$. A triangle is exactly half of a rectangle with the same base and height." — always check your work against this rule before you submit.*
8. **Try It 1:** A. 42 sq in — *Use the perpendicular height (7 in), not a slanted side. $A = \frac{1}{2} \times 12 \times 7 = \frac{1}{2} \times 84 = 42$ square inches.*
9. **Try It 2:** A. 27 sq ft — $A = \frac{1}{2} \times 6 \times 9 = \frac{1}{2} \times 54 = 27$ square feet.
10. **Exit Ticket:** A. 44 sq in — $A = \frac{1}{2} \times 11 \times 8 = \frac{1}{2} \times 88 = 44$ square inches.